

Econ 501

Homework 1 – Due October 8th

1. Assume there are only two goods, X and Y. Prove that quasi-concave utility $U(X,Y)$ is identical to $dMRX/dX < 0$
2. $U = XY^3$. Find the demands for X and Y
3. $U = X + 2Y$; $P_x = 2$; $P_y = 5$; $I = 50$. How much of X and Y should you buy?
4. $U = \min(2X, Y)$. Find the demands for X and Y
5. $U = X + \ln(Y)$. Find the demands for X and Y.
6. For each of the following utility functions, check for diminishing MRS.
 - a. $U = XY$
 - b. $U = \ln X + \ln Y$
 - c. $U = X^2 + Y^2$
 - d. $U = \ln X + Y$
 - e. $U = \min(X, Y)$

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Homework 2 – Due November 14th

Consider the following utility functions:

- a) $U = (xy)^2$
- b) $U = (xy)^{1/2}$
- c) $U = \min(x, 2y)$
- d) $U = 2x + y$
- e) $U = x^2y^2 + xy$

1. Check if any of the utility functions a) – e) are homogeneous. If so, state the degree of homogeneity of each.
2. All homogeneous utility functions are homothetic. Are any of the above functions homothetic but not homogeneous? Show your work.
3. Derive the Marshallian demands and indirect utility functions for a) – e).
4. All functions except c) are differentiable. Do these functions exhibit diminishing marginal utility? Are their Marshallian demands downward sloping? What can you infer about the necessity of diminishing marginal utility for downward-sloping demands?
5. Find the expenditure function and the two Hicksian demands for utility functions b) and c).

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Homework 3 – Due November 31th, 2024

Questions 1 – 6: Consider the following production functions:

a. $q = K^{1/2}L^{1/2}$ b. $q = 2K + L$ c. $q = \frac{4KL}{(K+L)}$ d. $q = KL$ e. $q = \min(K, \frac{1}{2}L)$

1. Find the returns to scale of each production function.
2. Find the value of, or the function for, the elasticity of substitution for each production function.
3. Suppose in the short run, capital is fixed at $\bar{K} = 64$, find the short-run production and cost function for each.

capital is no longer fixed

4. Find the equation for the firm's expansion path.
5. Assume $w = 2$ and $v = 8$, find the cost minimizing input mix to produce $q = 100$.
6. Find the long-run cost functions.

1. The domestic demand for color printers is $Q = 150,000 - 100P$. The domestic supply is

$$Q = 150P - 30,000$$

- Suppose no international trade takes place. Find the domestic autarky price and quantity.
- Suppose this country can import or export as many printers as it wants at the world price of $P_w = 600$. Would the country import or export printers? How many would it import/export under free trade?
- Suppose the domestic government puts a tariff of \$50 per imported printer. What would be the new market equilibrium (i.e., P ; domestic q_d and q_s ; and imports)? How much tariff revenue would the government gain?
- Assuming that supply = MC, what are the welfare effects of the tariff (i.e., ΔCS ; ΔPS ; ΔW)?
- Assume instead that the exporting nation signs a Voluntary Export Restraint with the domestic government that limits exports to 20,000 printers. Find the welfare effects of the VER as compared to free trade.

2. An exchange economy consists of two people, A and B, and two goods X and Y. A's preferences are represented by the utility function $U_A = X_A(Y_A)^2$; B's preferences are represented by the utility function $U_B = (X_B)^2 Y_B$. Finally, the initial endowments are as follow:

$$X_A^0 = 100; Y_A^0 = 50; X_B^0 = 200; Y_B^0 = 100.$$

- Draw the initial allocation in an Edgeworth box. Draw each person's indifference curve through the initial endowment.
- Is the initial allocation Pareto efficient? Please show your work - a yes or no answer will receive no credit even if correct.
- If your answer to part A is no, what type of trade do you expect to see between the two agents?
- What is the equation of the contract curve (write your answer as Y_A as a function of X_A)?